

Line Graphs of Trees

We searched for trees T such that the chromatic symmetric function of the line graph $L(T)$ fails to be e -positive. Using the tree data in `treesn.g6`, the first counterexample occurs on 7 vertices.

More precisely, consider the tree with edge set

$$\{\{0, 3\}, \{1, 4\}, \{2, 5\}, \{0, 6\}, \{1, 6\}, \{2, 6\}\}.$$

This is the 3-legged spider with center 6 and each leg of length 2. Its line graph has edge set

$$\{\{0, 3\}, \{1, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\}, \{4, 5\}\}.$$

For this graph,

$$X_{L(T)} = 6e_{3,2,1} - 6e_{3,3} + 6e_{4,1,1} + 12e_{4,2} + 18e_{5,1} + 12e_6.$$

In particular,

$$[e_{3,3}] X_{L(T)} = -6,$$

so $X_{L(T)}$ is not e -positive.

Its Schur expansion is

$$X_{L(T)} = 48s_{1,1,1,1,1,1} + 48s_{2,1,1,1,1} + 24s_{2,2,1,1} + 12s_{3,1,1,1} + 6s_{3,2,1}.$$

We also computed the chromatic symmetric functions of the line graphs $L(K_{2,n})$ for small n . In the cases below, all coefficients in the e -basis are positive. The last column records the sum of the coefficients in the e -expansion.

n	$X_{L(K_{2,n})}$ in the e -basis	sum of e -coefficients
1	$2e_2$	2
2	$2e_{2,2} + 12e_4$	14
3	$12e_{3,3} + 12e_{4,2} + 24e_{5,1} + 156e_6$	204
4	$216e_{4,4} + 96e_{5,3} + 336e_{6,2} + 864e_{7,1} + 3504e_8$	5016
5	$5280e_{5,5} + 2160e_{6,4} + 5760e_{7,3} + 14640e_{8,2} + 37440e_{9,1} + 120240e_{10}$	185520